8th Grade Physical Science

Lever Balance

Conservation of energy for an ideal machine:

$$F_1 d_1 = F_2 d_2 \quad \Rightarrow \quad m_1 g d_1 = m_2 g d_2$$

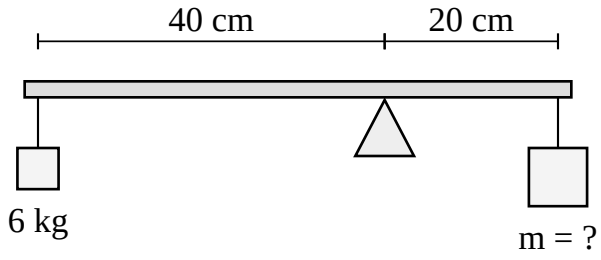
$$\frac{m_1 g d_1}{g} = \frac{m_2 g d_2}{g} \quad \Rightarrow \quad m_1 d_1 = m_2 d_2$$

Divide by g, the gravitational acceleration. It is the same on both sides.

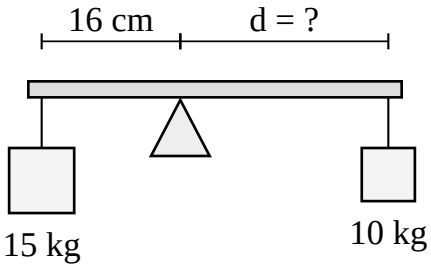
For levers, use distances from the fulcrum; the shared rotation factor cancels.

Massless beams. Diagrams drawn to scale.

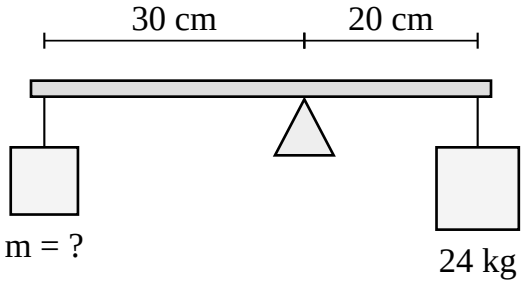
1. Find the missing mass.



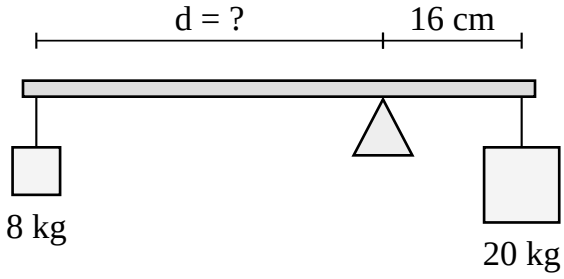
2. Find the missing distance.



3. Find the missing mass.



4. Find the missing distance.



Ramp Balance

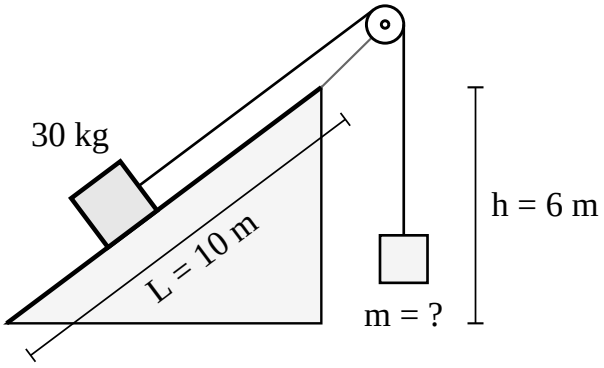
$$m_1 d_1 = m_2 d_2$$

$$m_{\text{hanging}}\, L = m_{\text{on ramp}}\, h$$

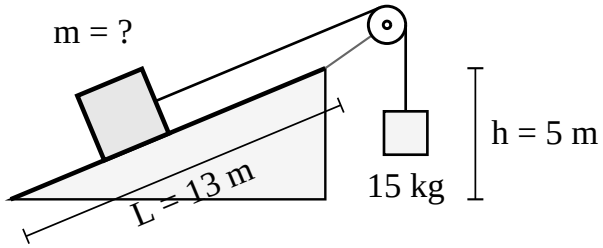
L = diagonal ramp length h = vertical height

Frictionless ramps; ideal ropes and pulleys. Diagrams drawn to scale.

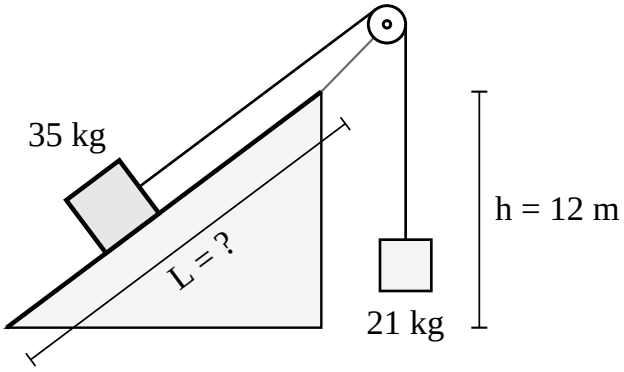
5. Find the missing mass.



6. Find the missing mass.



7. Find the ramp length.



8. Find the ramp length.

